

$R(A, B, C, D, E, F, H)$

FD:

$A \rightarrow D$

$AE \rightarrow H$

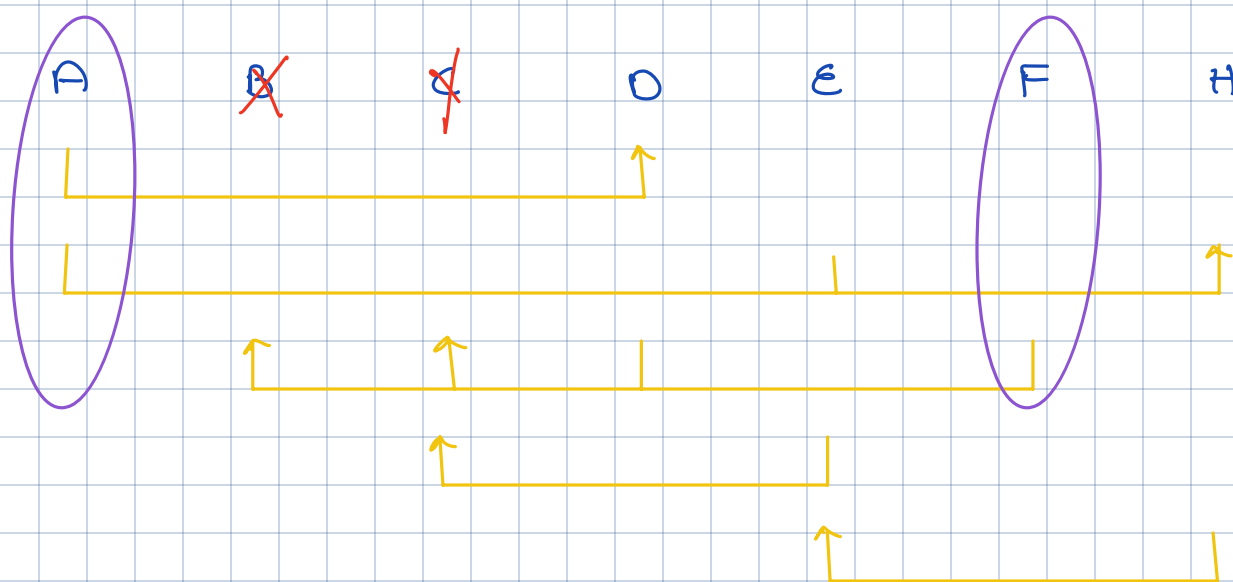
$DF \rightarrow BC$

$E \rightarrow C$

$H \rightarrow E$

↳ Those with no incoming arrows are always part of CK

↳ Those NOT on left-side of functional dependencies will never be part of CK



$(AF)^+ = ABCDF$ ✗

$(AEF)^+ = ABCDEFH$ ✓

$(AFH)^+ = ABCDEFH$ ✓

$(ADF)^+ = ABCDF$ ✗

$CK = \{AFH, AEF\}$

↳ Those attributes that are in CK are prime & others are non-prime attributes.

$\underbrace{AFH}_{CK} \rightarrow E \Rightarrow$ full functional dependency

$\underbrace{A}_{\text{A part of CK}} \rightarrow E \rightarrow$ Partial functional dependency

$B \rightarrow C$ $\rightarrow$ Transitive functional Dependency
 non-prime non-prime

Normalization:

1. 1NF:

$\hookrightarrow$ Schema on it's own is by default in this form
 $\hookrightarrow$ When we have data, then we can check & change it in 1NF

$\hookrightarrow$ 1NF disallows:

1. Multivalue attributes
2. Composite attributes

$\hookrightarrow$ Domain of attributes must be Atomic / Single value.

(a)

DEPARTMENT			
DNAME	<u>DNUMBER</u>	DMGRSSN	DLOCATIONS

$\hookrightarrow$ In 1NF

(b)

DEPARTMENT			
DNAME	<u>DNUMBER</u>	DMGRSSN	DLOCATIONS
Research	5	333445555	{Bellaire, Sugarland, Houston}
Administration	4	987654321	{Stafford}
Headquarters	1	888665555	{Houston}

$\hookrightarrow$ NOT in 1NF
 $\hookrightarrow$ Has multi-value

(c)

DEPARTMENT			
DNAME	<u>DNUMBER</u>	DMGRSSN	<u>DLOCATION</u>
Research	5	333445555	Bellaire
Research	5	333445555	Sugarland
Research	5	333445555	Houston
Administration	4	987654321	Stafford
Headquarters	1	888665555	Houston

Department

Name	DNo	DMGRSSN
Research	5	333445555
Research	5	333445555
Research	5	333445555
Administration	4	987654321
Headquarters	1	888665555

Locations

Dno
5
5
5
4
1

Dlocation

Bellaire
 Sugarland
 Houston
 Stafford
 Houston

$\hookrightarrow$ Is in 1NF

2. 2NF

$\hookrightarrow$ Must be already in 1NF

$\hookrightarrow$ There must be no Partial functional Dependency

Partial FD: Attribute(s) that is part of CK $\rightarrow$ Non-prime Attribute

Example,

$R(A, B, C, D, E, F)$

$AB \rightarrow C$

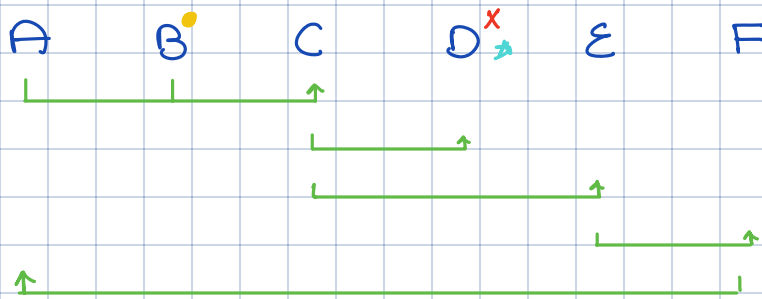
$C \rightarrow D$

$C \rightarrow E$

$E \rightarrow F$

$F \rightarrow A$

1. Find All CKs



* Even tho D can be part of CK but sir has done it like this. So shall we for understanding the Norm Process.

$B^+ : B$ X

$AB^+ : ABCDEF$ ✓

$BC^+ : ABCDEF$ ✓

$BD^+ : BD$ X

$BE^+ : ABCDEF$ ✓

$BF^+ : ABCDEF$ ✓

So, CK: $\{AB, BC, BE, BF\}$

$R(A, B, C) \quad A \rightarrow B, C \rightarrow D$

$A^+ : AB$

$C^+ : CD$

$R_1(A, B)$

$R_2(C, D)$

$R_3(A, C)$

put superkey of both in R_3

2. Identify PFD

$AB \rightarrow C$ X

$C \rightarrow D$ ✓

$C \rightarrow E$ X

$E \rightarrow F$ X

$F \rightarrow A$ X

3. Remove it

$\hookrightarrow$ Break the table on left-side & Remove Right side into

separate table

$R(A, B, C, D, E, F)$

$R(A, B, C, E, F)$

$R(C, D)$

$AB \rightarrow C$

$C \rightarrow E$

$E \rightarrow F$

$F \rightarrow A$

$C \rightarrow D$

3NF:

↳ Already in 2NF

↳ Must NOT have transitive Functional dependency

↳ It is 3NF when all FD have form:

LHS = Attribute(s) that is part of CK OR RHS = prime Attribute

↳ Since above is already in 3NF, we shall add attribute h FD:

Example,

$R(A, B, C, D, E, F, h)$

$AB \rightarrow C$

$C \rightarrow D$

$C \rightarrow E$

$E \rightarrow F$

$F \rightarrow A$

$C \rightarrow h$

$D \rightarrow h$

2NF:

$R(A, B, C, E, F, h)$ $R(C, D, h)$

$AB \rightarrow C$

$C \rightarrow E$

$E \rightarrow F$

$F \rightarrow A$

$C \rightarrow D$

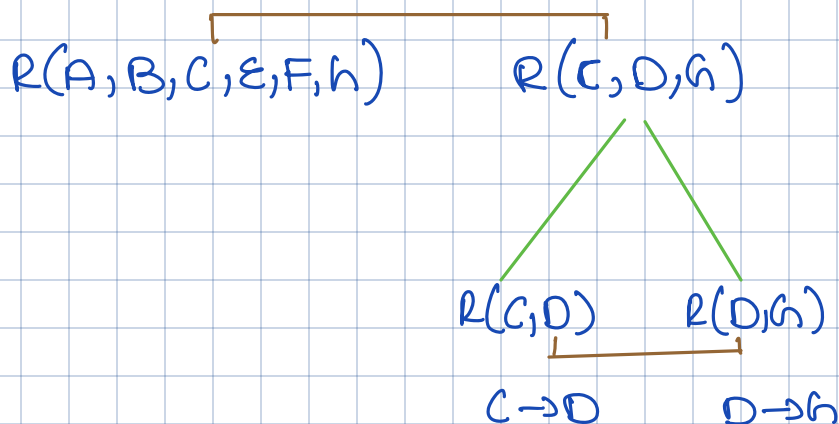
$C \rightarrow h$

$D \rightarrow h$

2. Identify TFD:

$AB \rightarrow C$ $\times$
 $C \rightarrow D$ $\times$
 $C \rightarrow E$ $\times$
 $E \rightarrow F$ $\times$
 $F \rightarrow A$ $\times$
 $C \rightarrow h$ $\times$
 $D \rightarrow h$ $\checkmark$

CK: $\{AB, BC, BE, BF\}$



Boyce - Codd NF:

$\hookrightarrow$ Left-side of each FD must be PK

Example,

OID	Odate	CID	CName	CState	PID	PDesc	Price	Qty
-----	-------	-----	-------	--------	-----	-------	-------	-----

$OID \rightarrow Odate, CID, CName, CState$

$CID \rightarrow CName, CState$

$PID \rightarrow PDesc, Price$

$OID, PID \rightarrow Qty$

CK = OID, PID

1. 2nd Normal form:

$OID \rightarrow Odate, CID, CName, CState$ $\checkmark$

$CID \rightarrow CName, CState$ $\times$

$PID \rightarrow PDesc, Price$ ✓

$OID, PID \rightarrow Qty$ ✗

$R(OID, Odate, CID, CName, CState, PID, PDesc, Price, Qty)$

$R(OID, Odate, CID, CName, CState)$

$OID \rightarrow Odate, CID, CName, CState$ ✗

$CID \rightarrow CName, CState$ ✓

$R(OID, PID, Qty)$ ✓

$R(PID, PDesc, Price)$ ✓

$OID, PID \rightarrow Qty$ ✗

$PID \rightarrow PDesc, Price$ ✗

3NF:

$R(OID, Odate, CID, CName, CState)$

$R(OID, Odate, CID)$ ✓

$OID \rightarrow Odate, CID$ ✓

$R(CID, CName, CState)$ ✓

$CID \rightarrow CName, CState$ ✓

In 3NF

In BCNF.

4NF:

- ↳ In BCNF
- ↳ No multi-value dependency

Example,

- ↳ Student has multiple skills

$SID \twoheadrightarrow Skills$

- ↳ Student can speak multiple languages

$SID \twoheadrightarrow Language$

Decompose

$R_1(SID, Skills)$ $R_2(SID, Language)$

5NF:

- ↳ Is in 4NF
- ↳ Has no Join dependency

A relation that has a join dependency can not be divided into two (or more) relations such that the resulting tables can be natural joined to form the original table.

R
 $R_1 \quad R_2 \quad R_3$

Natural Join
in any order

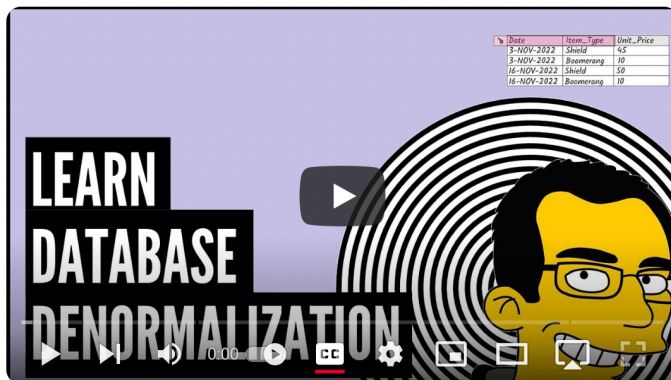
$R_1 * (R_2 * R_3)$ or $(R_1 * R_2) * R_3$

↳ you get R back then R is in 5NF

5NF:

If a table can be thought of as a logical result of joining other tables

Denormalization:



Learn Database Denormalization



Decomplexify

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1.9K



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What is RDBMS denormalization all about? This video will help you to recognize

Q20.

Emp-dep

1. Insert anomaly $\rightarrow$ Have to insert emp before Department
2. Update anomaly $\rightarrow$ Changing name of Dep changes all occurrences
3. Delete anomaly $\rightarrow$ Deleting Borg, James E deletes HQ dep

Emp-proj

1. Insert anomaly $\rightarrow$ Add emp before proj
2. Update anomaly $\rightarrow$ Changing a attribute requires changes in all tuples
3. Delete anomaly $\rightarrow$ If only one emp works on proj then deleting him deletes the proj

Q24.

$R(A, B, C, D, E, F, G, H, I, J)$

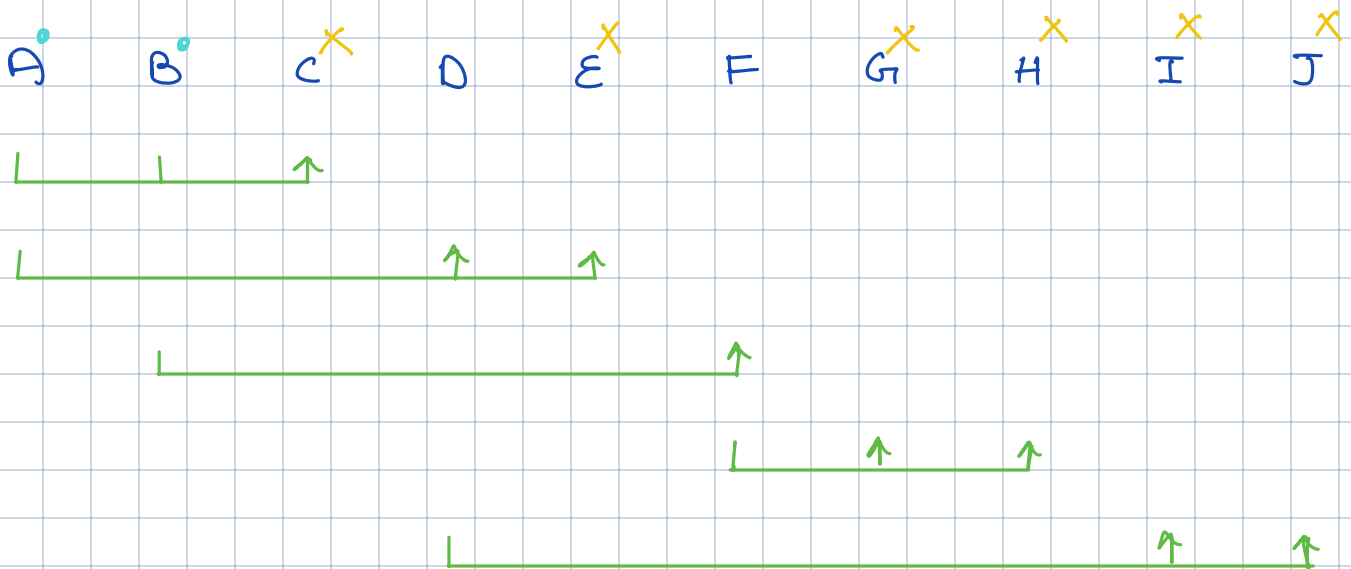
$AB \rightarrow C$

$A \rightarrow DE$

$B \rightarrow F$

$F \rightarrow GH$

$D \rightarrow IJ$



$(AB)^+$: ABCDEFGHIJ

1. 1NF
Yes

2. 2NF:

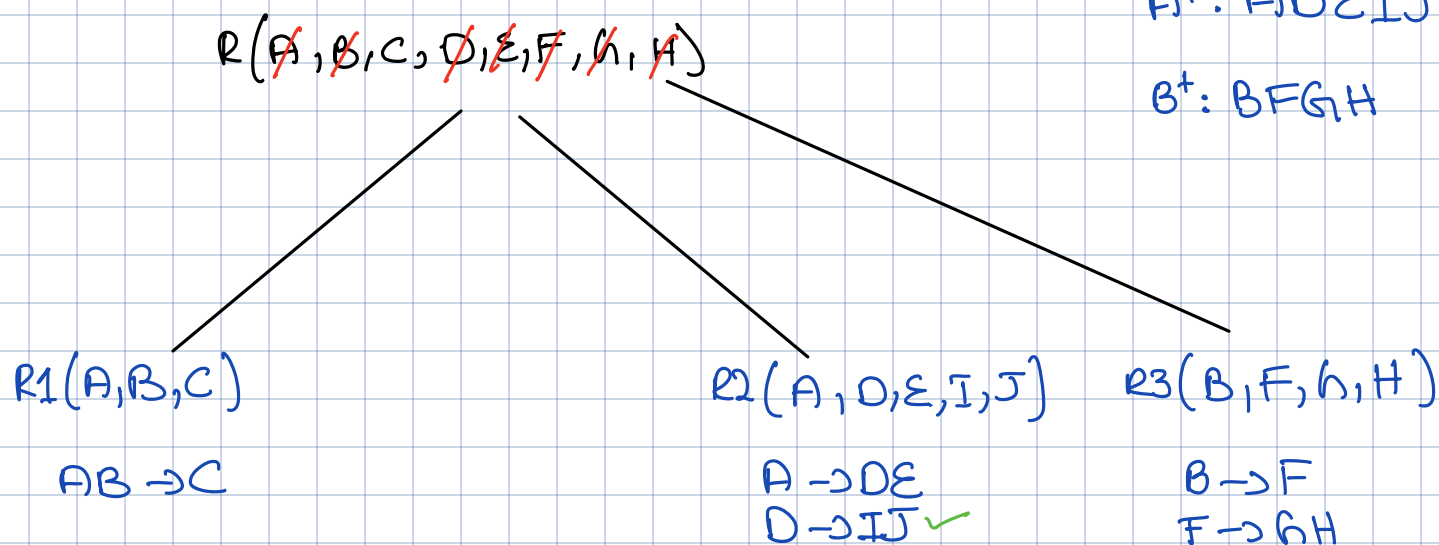
Part of CK $\rightarrow$ non-prime attributes

Check
down

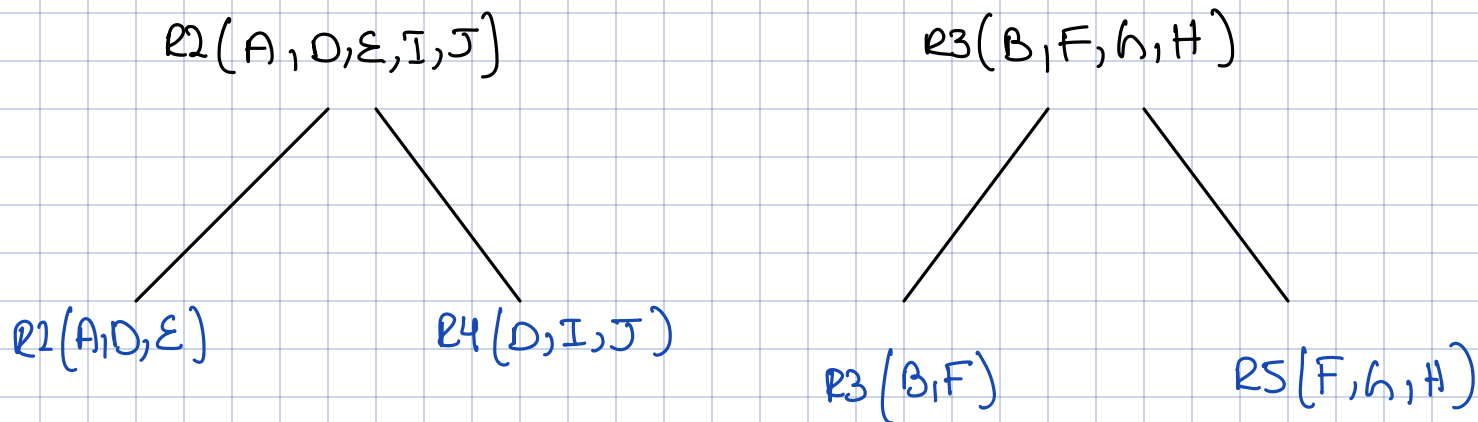
$AB \rightarrow C$ $\times$
 $A \rightarrow DE$ $\checkmark$
 $B \rightarrow F$ $\checkmark$
 $F \rightarrow GH$ $\times$
 $D \rightarrow IJ$ $\times$

A^+ : ADEIJ

B^+ : BFGH



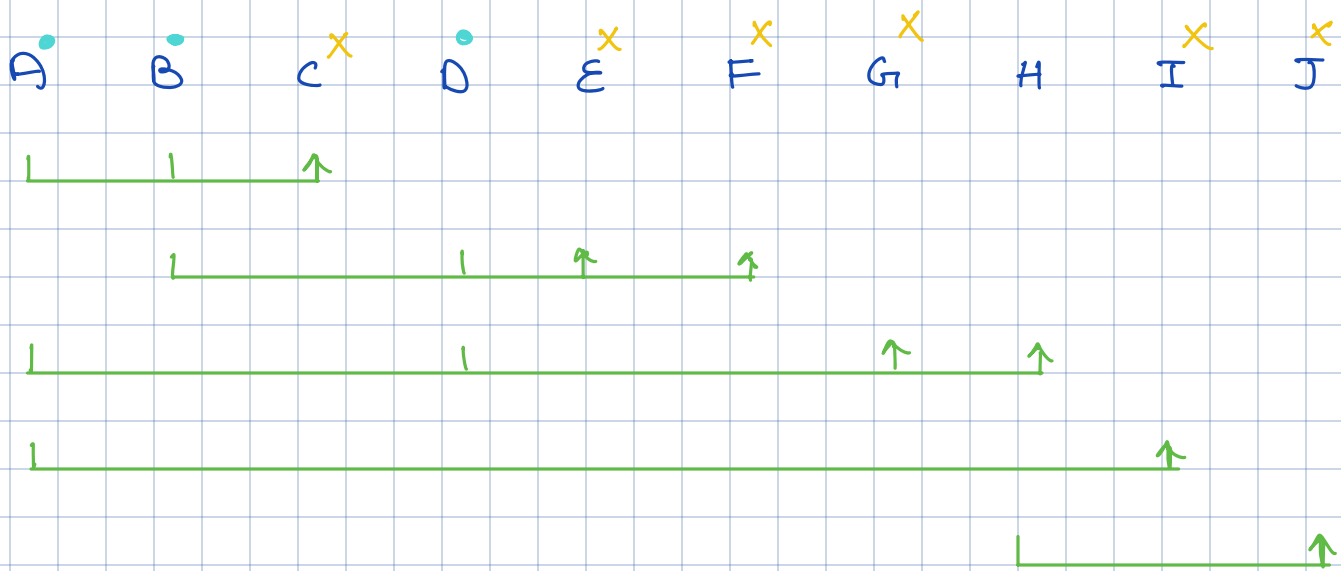
3. 3NF:



Q25.

$AB \rightarrow C$
 $BD \rightarrow EF$
 $AD \rightarrow GH$
 $A \rightarrow I$

$H \rightarrow J$



$(ABD)^+ : ABCDEFGHIJ$

1. 1NF
Yes!

$AB^+ : ABC$

$BD^+ : BDEF$

$AD^+ : ADGHIJ$

$A^+ : AI$

A, B, D

Check
down

2. 2NF:

$AB \rightarrow C$ ✓
 $BD \rightarrow EF$ ✓
 $AD \rightarrow GH$ ✓
 $A \rightarrow I$ ✓
 $H \rightarrow J$

$R(A, B, D)$

$R_1(A, B, C)$

$AB \rightarrow C$

$R_2(B, D, E, F)$

$BD \rightarrow EF$

$R_4(A, I)$

$A \rightarrow I$

$R_3(A, D, G, H, J)$

$AD \rightarrow GH$
 $H \rightarrow J$ ✓

3. 3NF

$R_3(A, D, G, H)$ $R_4(H, J)$

Q28.

R	A	B	C X	D X	E X	F X	G ●	H ●	I	J	X K
Course_no	Sec_no	Offering_dept	Credit_hours	Course_level	Instructor_ssn	Semester	Year	Days_hours	Room_no	No_of_students	
FD1			↑	↑	↑						
FD2					↑				↑	↑	↑
FD3	↑	↑			↑						

GH^+ : GH

ABGH: ABGH CDE IJ K F ✓

AGH^+ : AGH CDE

AGHI: AGHI CDE

BGH^+ : BGH

AGHJ: AGHJ CDE

GHI: GHI

BGHI: BGHI

GHT: GHT

BGHT: BGHT

GHIJ: GHIJ F A B CDE K ✓

Check
down

CK = Semester, Year, Course-no, Sec-no

↓

Semester, Year, Days-hours, Room-no

1. 2NF:

$Course_no^+$: Offeringdept, Credit-hours, Course-level, Course-no

R1(Course-no, offering-dept, Credit-hours, Course-level)

R2(Course-no, Sec-no, Instructor-ssn, Semester, Year, Days-hours, Room-no, No-of-students)

Q29-

ORDER (O#, Odate, Cust#, Total-amount)

ORDER-ITEM (O#, I#, Qty-ordered, Total-price, Discount%)

R (O#, I#, Cust^x#, Total^x-amount, Qty^x-ordered, Total^x-price, Dist^x%)

O# → Odate, Cust#, Total-amount ✓

O#, I# → Qty-ordered, Total-price, Discount%

Discount → I#

Qty-ordered, Total-price, Discount% → Total-amount

O# : O#, Odate, Cust#, Total-amount ✗

O# I# : O# I# Odate Cust# Total-amount, Qty-ordered, total-price, discount% ✓

Not in 2NF, 3NF

Check down

Q30-

Cas-sale (Cas#, Date-sold, Saleperson, Commission%, Dist-amt)

Date-sold → Dist-amt ✓

Saleperson → Commission% ✓

In 1NF

NOT in 2NF, 3NF

Check down

1NF:

Saleperson⁺: Saleperson, Commission

R1 (Car#, Date-sold, Salesperson, Discount-amount)

R2 (Salesperson, Commission)

2NF:

R1 (Car#, Date-sold, Salesperson)

R2 (Salesperson, Commission)

R3 (Date-sold, Discount-amount)

Q31.

Book (Book-title, Author-name, Book-type, List-price, Author-affil, Publisher)

Book-title $\rightarrow$ Publisher, Book-type ✓

Book-type $\rightarrow$ List-price ✓

Author-name $\rightarrow$ Author-affil ✓

Check down

CK = Book-title, Author-name

(a) 1NF

(b) Book-title⁺: Book-name, Publisher, book-type, List-price

Author-name⁺: Author-name, Author-affil

2NF

R1 (Book-title, Publisher, book-type, List-price)

R2 (Author-name, Author-affil)

R3 (Book-title, Author-name)

3NF:

R1 (Book-title, Publisher, book-type)

R2 (Book-type, List-price)

R2 (Author-name, Author-affil)

R3 (Book-title, Author-name)

Q32.

(a) Manufacturer, serial-number $\rightarrow$ model, ..., Retailer

(b) model $\rightarrow$ manufacturer

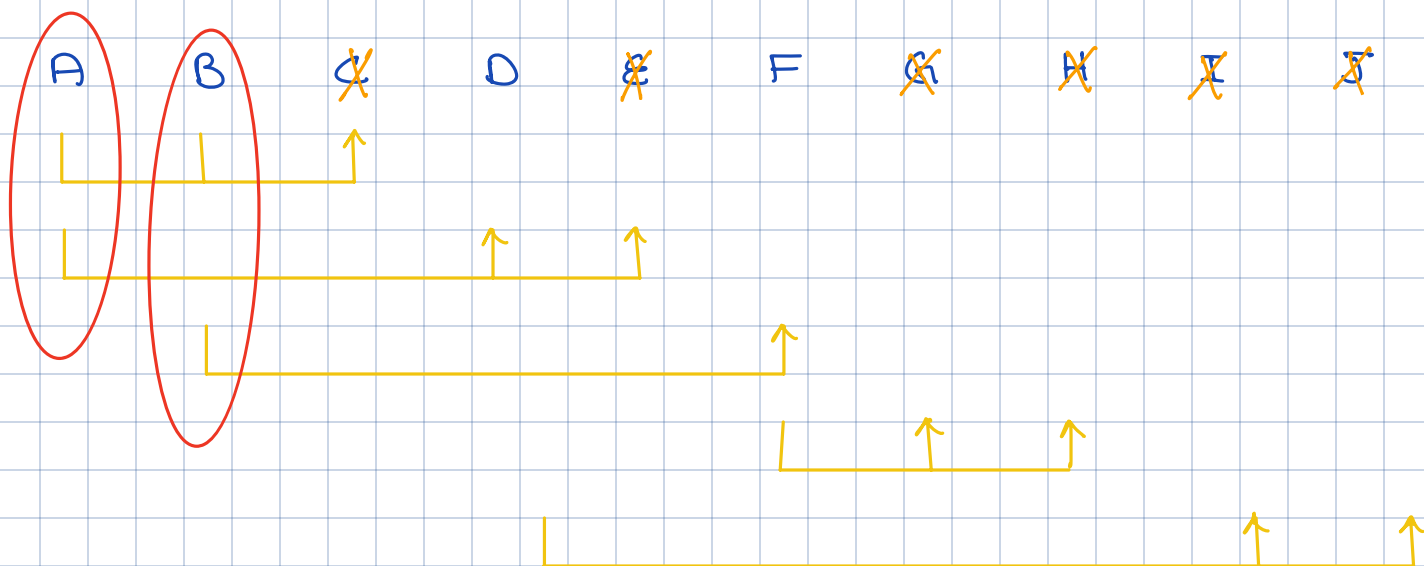
(c) batch $\rightarrow$ model

(d) model, manufacturer $\rightarrow$ capacity

Q24.

$R(A, B, C, D, E, F, G, H, I, J)$

$AB \rightarrow C$
 $A \rightarrow DE$
 $B \rightarrow F$
 $F \rightarrow GH$
 $D \rightarrow IJ$



AB^+ : ABCDEFGHIJ

CK: $\{AB\}$ Prime attributes: $\{A, B\}$ Non-Prime attribute: $\{C, D, E, F, G, H, I, J\}$

2NF:

$AB \rightarrow C$
 $A \rightarrow DE$ ✓
 $B \rightarrow F$ ✓
 $F \rightarrow GH$
 $D \rightarrow IJ$

$R_1(A, D, E, I, J)$

$R_2(B, F, G, H)$

$R_3(A, B, C)$

$A \rightarrow DE$

$B \rightarrow F$
 $F \rightarrow GH$

$AB \rightarrow C$

3NF:

$R_3(F, G, H)$ $R_4(D, I, J)$ $R(A, B, C)$ $R_1(A, D, E)$ $R_2(B, F)$

Q25.

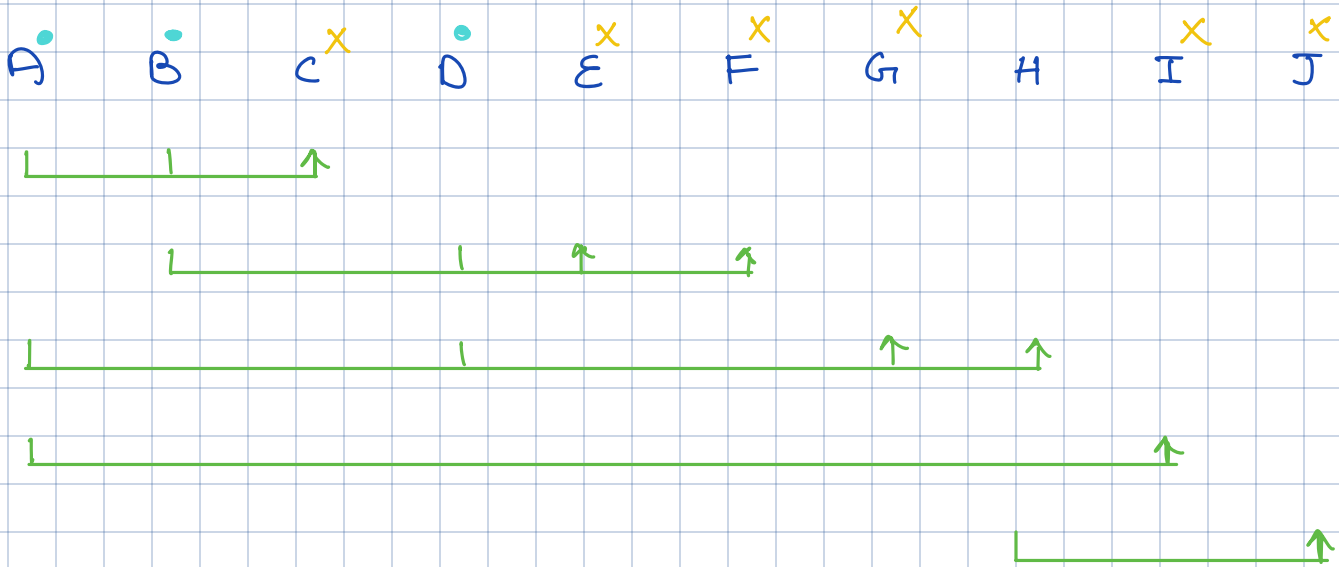
$AB \rightarrow C$

$BD \rightarrow EF$

$AD \rightarrow GH$

$A \rightarrow I$

$H \rightarrow J$



$(ABD)^+ : \cancel{A} \cancel{B} \cancel{C} \cancel{D} \cancel{E} \cancel{F} \cancel{G} \cancel{H} \cancel{I} \cancel{J}$ CK: $\{ABD\}$ PA: $\{A, B, D\}$ NPA: $\{C, E, F, G, H, I, J\}$

2NF:

$AB \rightarrow C$ ✓

$BD \rightarrow EF$ ✓

$AD \rightarrow GH$ ✓

$A \rightarrow I$ ✓

$H \rightarrow J$

$R_1(A, B, C)$

$R_2(B, D, E, F)$

$R_3(A, D, G, H, J)$

$R_4(A, I)$

$R_5(A, D, B, J) \rightarrow H \rightarrow J$, lost

move J to where H is.

3NF:

$H \rightarrow J$ ✓
X X

$R_6(H, J)$ $R_3(A, D, G, H)$

Q28.

R	A	B	C X	D X	E X	F X	G •	H •	I	J	X	K
	Course_no	Sec_no	Offering_dept	Credit_hours	Course_level	Instructor_ssn	Semester	Year	Days_hours	Room_no	No_of_students	
FD1			↑	↑	↑							
FD2	↑	↑				↑	↑	↑	↑	↑	↑	↑
FD3	↑	↑				↑	↑	↑	↑	↑	↑	↑

GH^+ : GH

$ABGH$: $ABGH CDE IJ K F$ ✓

AGH^+ : $AGH CDE$

$AGHI$: $AGHI CDE$

BGH^+ : BGH

$AGHJ$: $AGHJ CDE$

GHI : GHI

$BGHI$: $BGHI$

$GHIJ$: $GHIJ$

$BGHIJ$: $BGHIJ$

$GHIJ$: $GHIJ F A B CDE K$ ✓

$CK = \text{Semester, Year, Course-no, Sec-no}$

↓

$\text{Semester, Year, Days-hours, Room-no}$

$PA: \{ \text{Semester, Year, Course-no, Sec-no, Days-hours, Room-no} \}$

$NPA: \{ \text{Offering-dept, Credit-hours, Course-level, Instructor-ssn} \}$

2NF:

{Course_no} → {Offering_dept, Credit_hours, Course_level} ✓

{Course_no, Sec_no, Semester, Year} → {Days_hours, Room_no,
No_of_students, Instructor_ssn}

{Room_no, Days_hours, Semester, Year} → {Instructor_ssn, Course_no,
Sec_no}

$R_1(\text{Course_no}, \text{Offering_dept}, \text{Credit_hours}, \text{Course_level})$

{Course_no} → {Offering_dept, Credit_hours, Course_level} ✗

$R_2(\text{Course_no}, \text{Sec_no}, \text{Instructor_ssn}, \text{Semester}, \text{Year}, \text{Days_hours},$
 $\text{Room_no}, \text{No_of_students})$

{Course_no, Sec_no, Semester, Year} → {Days_hours, Room_no,
No_of_students, Instructor_ssn}

{Room_no, Days_hours, Semester, Year} → {Instructor_ssn, Course_no,
Sec_no}

3NF

Already in 3NF as well

Q30.

CAR_SALE(Car#, Date_sold, Salesperson#, Commission%, Discount_amt)

Date_sold → Discount_amt and

Salesperson# → Commission%

2NF:

Date_sold → Discount_amt and

Salesperson# → Commission% ✓

$R_1(\text{Salesperson\#}, \text{Commission\%})$

Salesperson# → Commission%

$R_2(\text{Car\#}, \text{Date_sold}, \text{Salesperson\#}, \text{Discount_amt})$

Date_sold → Discount_amt TFD

3NF:

$R_3(\text{Date_sold}, \text{Discount_amt})$

Date_sold → Discount_amt

$R_2(\text{Car\#}, \text{Date_sold}, \text{Salesperson\#})$

$R_1(\text{Salesperson\#}, \text{Commission\%})$

$\text{Salesperson\#} \rightarrow \text{Commission\%}$

Q31.

BOOK (Book_title, Author_name, Book_type, List_price, Author_affil, Publisher)

$\text{Book_title} \rightarrow \text{Publisher, Book_type}$

$\text{Book_type} \rightarrow \text{List_price}$

$\text{Author_name} \rightarrow \text{Author_affil}$

CK = Book-title, Author-name

2NF:

$\text{Book_title} \rightarrow \text{Publisher, Book_type}$ ✓ PFD

$\text{Book_type} \rightarrow \text{List_price}$

$\text{Author_name} \rightarrow \text{Author_affil}$ ✓ PFD

$R_1(\text{Book_title, Publisher, Book_type, List_price})$

$\text{Book_title} \rightarrow \text{Publisher, Book_type}$ ✗

$\text{Book_type} \rightarrow \text{List_price}$ ✓

$R_2(\text{Author_name, Author_affil})$

$\text{Author_name} \rightarrow \text{Author_affil}$ ✗

$R_3(\text{Book_title, Author_name})$

3NF:

$R_{1A}(\text{Book_title, Publisher, Book_type})$

$\text{Book_title} \rightarrow \text{Publisher, Book_type}$

$R_{1B}(\text{List_price, Book_type})$

$\text{Book_type} \rightarrow \text{List_price}$

$R_2(\text{Author_name}, \text{Author_affil})$

$\text{Author_name} \rightarrow \text{Author_affil}$

Q33.

$R(\text{Doctor\#}, \text{Patient\#}, \text{Date}, \text{Diagnosis}, \text{Treat_code}, \text{Charge})$

$\text{Doctor\#}, \text{Patient\#} \rightarrow \text{Diagnosis}$

$\text{Treat_code} \rightarrow \text{Charge}$

ck: $\{\text{Doctor\#}, \text{Patient\#}, \text{Treat_code}, \text{Date}\}$

2NF:

$\text{Doctor\#}, \text{Patient\#} \rightarrow \text{Diagnosis}$ ✓

$\text{Treat_code} \rightarrow \text{Charge}$ ✓

$R_1(\text{Doctor\#}, \text{Patient\#}, \text{Diagnosis})$

$\text{Doctor\#}, \text{Patient\#} \rightarrow \text{Diagnosis}$ ✓

$R_2(\text{Treat_code}, \text{Charge})$

$\text{Treat_code} \rightarrow \text{Charge}$ ✓

$R_3(\text{Doctor\#}, \text{Patient\#}, \text{Treat_code}, \text{Date})$

3NF:

Already in 3NF

Q35.

(c) $\text{Book_Name} \twoheadrightarrow \text{Author}$

$\text{Book_Name} \twoheadrightarrow \text{Edition, Year}$

$R_1(\text{Book_Name}, \text{Author})$

$R_2(\text{Book_Name}, \text{Edition}, \text{Year})$

Q36.

(a) Trip-id $\rightarrow$ Start_date

Trip-id $\rightarrow$ Cities-visited | Cards-used CK: {Trip-id, cities-visited, Cards-used}

2NF

Trip-id $\rightarrow$ Start_date ✓ PFD

Trip-id $\rightarrow$ Cities-visited | Cards-used

$R_1(\text{Trip-id}, \text{Start-date})$ $R_2(\text{Trip-id}, \text{Cities-visited}, \text{Cards-used})$

Trip-id $\rightarrow$ Start_date

Trip-id $\rightarrow$ Cities-visited | Cards-used

3NF:

Yes

BCNF:

Yes!

4NF:

Trip-id $\rightarrow$ Cities-visited | Cards-used ✓ NVD

$R_1(\text{Trip-id}, \text{Start-date})$

$R_2(\text{Trip-id}, \text{Cities-visited})$

$R_3(\text{Trip-id}, \text{Cards-used})$

Q29.

(O#, Odate, Cust#, Total-amount, I#, Qty-ordered, Total-price, Dist%)

O# $\rightarrow$ Odate, Cust#, Total-amount

I# $\rightarrow$ Qty-ordered, Total-price, Dist%.

Dist% $\rightarrow$ I#

Qty-ordered, Total-price, Discount% $\rightarrow$ Total-amount

O#, I#⁺: Odate, Cust#, Total-amount, Qty-ordered, Total-price, Dist%
O#, I# ✓

2NF:

O# $\rightarrow$ Odate, Cust#, Total-amount
✓ ✓ ✓

I# $\rightarrow$ Qty-ordered, Total-price, Dist%
✓ ✓ ✓

Dist% $\rightarrow$ I#
✗

Qty-ordered, Total-price, Discount% $\rightarrow$ Total-amount
✗

NOT in 2NF & by extension NOT in 3NF